

1 Problem Description

A cube and a sphere have the same volume and same centre, If $p\%$ of the surface area of the cube lies outside the sphere, what is p ?

2 Problem Solution

Let volume = v

Let side length = s

Let radius of sphere = r_{sphere}

Let radius of circle = r_{circle}

Then

$$\begin{aligned} s &= \sqrt[3]{v} \\ r_{sphere} &= \sqrt[3]{\frac{3v}{4\pi}} \\ &= v^{\frac{1}{3}} \left(\frac{3}{4\pi} \right)^{\frac{1}{3}} \end{aligned}$$

So find the area of the circle intersecting the cube face. The radius can be found by saying the adjacent equals $\frac{s}{2}$ and the hypotenuse equals r

$$\begin{aligned}
r_{circle}^2 &= r_{sphere}^2 - \frac{s^2}{2} \\
&= v^{\frac{2}{3}} \left(\frac{3}{4\pi} \right)^{\frac{2}{3}} - \frac{v^{\frac{2}{3}}}{4} \\
&= v^{\frac{2}{3}} \left(\left(\frac{3}{4\pi} \right)^{2/3} - \frac{1}{4} \right)
\end{aligned}$$

So surface area of one face equals $s^2 - r_{circle}^2 \pi$

$$\begin{aligned}
\text{So p} &= \frac{s^2 - r_{circle}^2 \pi}{s^2} \\
&= 1 - \frac{r_{circle}^2 \pi}{s^2} \\
&= 1 - \frac{v^{\frac{2}{3}} \left(\left(\frac{3}{4\pi} \right)^{2/3} - \frac{1}{4} \right) \pi}{v^{\frac{2}{3}}} \\
&= 1 - \left(\left(\frac{3}{4\pi} \right)^{2/3} - \frac{1}{4} \right) \pi \\
&= 0.576
\end{aligned}$$

This is all under the assumption that the circle can fully fit in the cubes face. Is this assumption true?

Well for it to be true,

$$\begin{aligned}
 r_{circle} &\leq \frac{s}{2} \\
 \implies 2\sqrt{v^{\frac{2}{3}}\left(\left(\frac{3}{4\pi}\right)^{2/3} - \frac{1}{4}\right)} &\leq v^{\frac{1}{3}} \\
 \implies 2v^{\frac{1}{3}}\sqrt{\left(\frac{3}{4\pi}\right)^{2/3} - \frac{1}{4}} &\leq v^{\frac{1}{3}} \\
 \implies 2\sqrt{\left(\frac{3}{4\pi}\right)^{2/3} - \frac{1}{4}} &\leq 1 \\
 &\text{which is true for all } v \text{ as } 0.734 < 1
 \end{aligned}$$

THE CLUE WAS 102